

Feng-Kai (Leonel) Huang

Ph.D. Student

AI Researcher

Department of Computer Science and Information Engineering

National Taiwan University

+886-922332909

✉ fengkaihuang95033@gmail.com

🌐 Personal Website

🎓 Feng-Kai Huang

📄 leohuang0511 🐙 LeoHuang0511

SUMMARY

Ph.D. student in Computer Science and Information Engineering at National Taiwan University and a two-time inductee of the Phi Tau Phi Scholastic Honor Society. Focused on computer vision, with specific expertise in crowd counting, video individual counting, multi-object tracking, and image editing, with publications in top-tier international conferences.

EDUCATION

National Taiwan University (NTU)

Ph.D. Student in Computer Science and Information Engineering

Aug. 2024 - Present

Taipei, Taiwan

- GPA: 4.22/4.3. Advisor: **Wen-Huang Cheng** & **Hong-Han Shuai**.
- Communications and Multimedia Laboratory (**CMLab**) - **AIMM** Research Group.
- Big data Analytics and Social Intelligent Computing (**BASIC**) Laboratory.

National Yang Ming Chiao Tung University (NYCU)

Master of The Graduate Degree Program of Artificial Intelligence

Sep. 2022 - Jul. 2024

Hsinchu, Taiwan

- GPA: 4.25/4.3. Advisor: **Hong-Han Shuai**.
- Big data Analytics and Social Intelligent Computing (**BASIC**) Laboratory.
- Honorary Member of The Phi Tau Phi Scholastic Honor Society of the Republic of China [\[Link\]](#).

Chung Yuan Christian University (CYCU)

Bachelor of Electrical Engineering

Sep. 2018 - Jun. 2022

Taoyuan, Taiwan

- GPA: 3.98/4. Rank#1.
- Honorary Member of The Phi Tau Phi Scholastic Honor Society of the Republic of China [\[Link\]](#).

PUBLICATIONS

Conference Papers

- **Flowing Crowd to Count Flows: A Self-Supervised Framework for Video Individual Counting**
Huang, F. K., Huang, B. L., Tsao, L. W., Wu, J. C., Shuai, H. H., Cheng, W. H.
ACM International Conference on Multimedia (MM), 2025. (Regular Track Paper) [\[Website\]](#) [\[Code\]](#)
- **OinkTrack: An Ultra-Long-Term Dataset for Multi-Object Tracking and Re-Identification of Group-Housed Pigs**
Huang, F. K., Xu, H. W., Lee, C. C., Tu, H. Y., Shuai, H. H., Cheng, W. H.
ACM International Conference on Multimedia (MM), 2025. (Dataset Track Paper) [\[Website\]](#) [\[Code\]](#)
- **When Anchors Meet Cold Diffusion: A Multi-Stage Approach to Lane Detection**
Huang, B. L., Ni, Z. X., **Huang, F. K.**, Shuai, H. H., Cheng, W. H.
International Conference on Computer Vision (ICCV), 2025. (Accepting rate = 24% (2698/11239)) [\[Code\]](#)
- **IterDiff: Training-Free Iterative Face Editing via Efficient CLIP-guided Memory Bank**
Chiu, C. Y., **Huang, F. K.**, Hsiao, T. F., Shuai, H. H., Cheng, W. H.
IEEE International Conference on Image Processing (ICIP), 2025. [\[Paper\]](#) [\[Code\]](#)
- **Density-Based Flow Mask Integration via Deformable Convolution for Video People Flux Estimation**
Wan, C. L., **Huang, F. K.**, Shuai, H. H.
IEEE Winter Conference on Application of Computer Vision (WACV), 2024. [\[Paper\]](#) [\[Code\]](#)
- **Smart Speaker Based on Detection of Millimeter Wave**
Liao, Y. P., **Huang, F. K.**, Xia, Y. J., Cheng, H.
IEEE International Conference on Consumer Electronics-Taiwan (ICCE-TW), 2022. [\[Paper\]](#)

Under Review

- **Density-Assisted Adaptive Alignment Network for Video Individual Counting**
Huang, F. K., Weng, K. J., Wu, J. C., Shuai, H. H., Cheng, W. H.
Under Review at ACM Transactions on Intelligent Systems and Technology (TIST), 2025.

HONORS & AWARDS

Academic Honors & Scholarship

- 2025 – **ACM SIGMM Student Travel Grant (ACM MM 2025)** [\[Link\]](#)
- 2024 – **Graduate Research Fellowship** of National Science and Technology Council [\[Link\]](#)
- 2023 – **Certificate of Excellence** (Spring 2022-2023) at NYCU [\[Link\]](#)
- 2018–2021 – **Certificate of Excellence** (Six semesters from 2018 to 2021) at CYCU [\[Link\]](#)
- 2018–2021 – **Subjects Merit Awards** (Calculus I, Circuit Theory I & II, Engineering Mathematics II, Electronics II, The Principles of Communications, Business Data Communication, Electromagnetism) at CYCU [\[Link\]](#)

International Competitions

- 2021 – **Gold Prize**, 2021 Seoul International Invention Fair [\[Link\]](#)

Domestic Competitions

- 2022 – **Honorable Mention Award**, 2022 AI CUP Competition on UAV Smart Counting [\[Link\]](#)
- 2021 – **Honorable Award**, 2021 Student Engineering Papers Competition of the Chinese Institute of Engineers [\[Link\]](#)
- 2021 – **1st Place (TQC+)** & **3rd Place (IP)**, International ICT Innovative Services Awards 2021 [\[Link\]](#)
- 2021 – **3rd Place (TQC+)**, International ICT Innovative Services Awards 2020 [\[Link\]](#)

PROJECTS

Multi-sensor AI Chips for Precision Swine Pathogen Detection

- Utilizing AI to track the activity levels and body temperatures of captive pigs and detect abnormalities, combined with a multi-antigen biosensor chip to identify unknown infectious diseases affecting pig populations.

FasterBID: Accelerating Bidirectional Decoding with Uncertainty Gating and Parameter-Free Weak Priors

- Accelerating the speed of Vision Language Action models during test-time adaptation, achieving a 1.6x speedup while maintaining accuracy through the design of uncertainty gating and fast-weak policies.

SKILLS

Programming: Python, C/C++, LaTeX, Verilog

Research Interest: Artificial Intelligence, Computer Vision, Multimedia Processing, Generative Models, Crowd Counting, Video Individual Counting, Multi-object Tracking

Developer Tools: Jetson Nano, Raspberry Pi, Arduino

Languages: Mandarin (native), English (**TOEIC** 800/990)

PROFESSIONAL ACTIVITIES

Conference Reviewer: ICASSP 2026, ACM MM 2025, ICME 2025, AVSS 2025, TAAI 2024

Journal Reviewer: IEEE TPAMI, ACM TOMM

WORK EXPERIENCE

National Taiwan University

Mar. 2025 - Dec. 2025

Teaching Assistant

Taipei, Taiwan

- Artificial Intelligence: co-designed three hands-on assignments covering MLLM, RAG, and Multi-Agent Systems for the Artificial Intelligence course.
- Computer Vision Practice with Deep Learning: designed two hands-on assignments covering object detection and image generation for Computer Vision Practice with Deep Learning.
- Assisted students in completing their assignments, and managed all assessment activities.

PATENTS

Rotate-type smart speaker and rotate-type home appliance controller, *Patent No. TWI817194B*, CYCU, 2023.

Liao, Y. P., **Huang, F. K.**, Xia, Y. J., Cheng, H. [\[Link\]](#)

System for detecting fecal and toilet lid with fecal detection thereof, *Patent No. TWI788834B*, CYCU, 2023

Huang, F. K., Xia, Y. J., Cheng, H. [\[Link\]](#)